

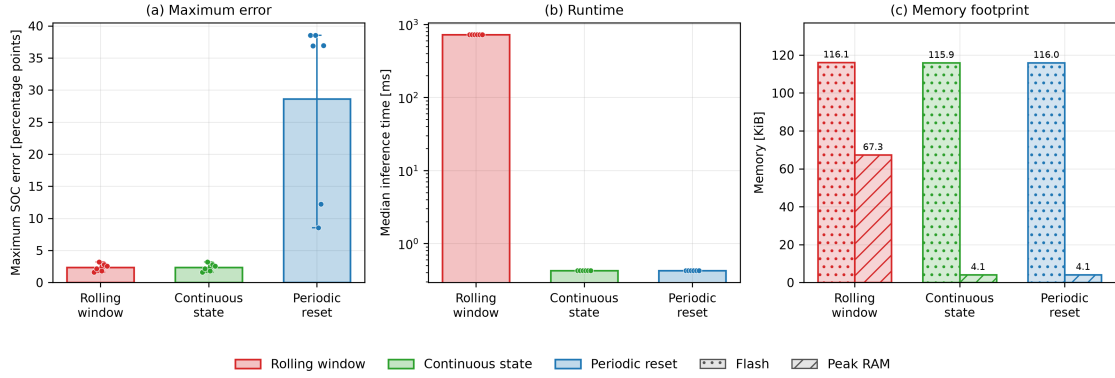


Figure 6.20: DD inference-mode trade-off on the STM32. The panels compare maximum dataset-SOC error, median latency, compiled flash occupancy, and measured peak runtime RAM for exact rolling-window, continuous-state, and periodically reset recurrent execution. Cell points show that periodic resets, rather than continuous forwarding, create the dominant accuracy loss.

6.5 Discussion

6.5.1 Structural interpretation across estimator classes

The four implementations expose distinct relationships between update mechanism, available correction information, and disturbance response. DM has the largest nominal error and reacts strongly to persistent current-gain and current-offset errors and to missing samples. Its current-integration structure has no voltage-residual correction. A practical full-charge anchor limits accumulated drift, but the lifecycle analysis shows that this anchor restructures the gain-error contribution without removing it uniformly. HDM adapts the capacity denominator through learned SOH and reduces capacity mismatch. It retains the same charge-integration core, so its responses to systematic current errors, missing charge information, and quantization remain structurally similar to DM.

HECM combines current-driven propagation with a voltage residual and has the strongest observed recovery profile under the tested initialization mapping. Additive current offset produces its largest global disturbance penalty. High current noise and burst dropout are the most consequential remaining cases because corrupted or frozen current affects state propagation and observer correction. The supplementary lookup analysis shows that the largest local lookup and disturbance interaction is 0.00612 MAE, compared with a 0.1763 HECM penalty from the ad-